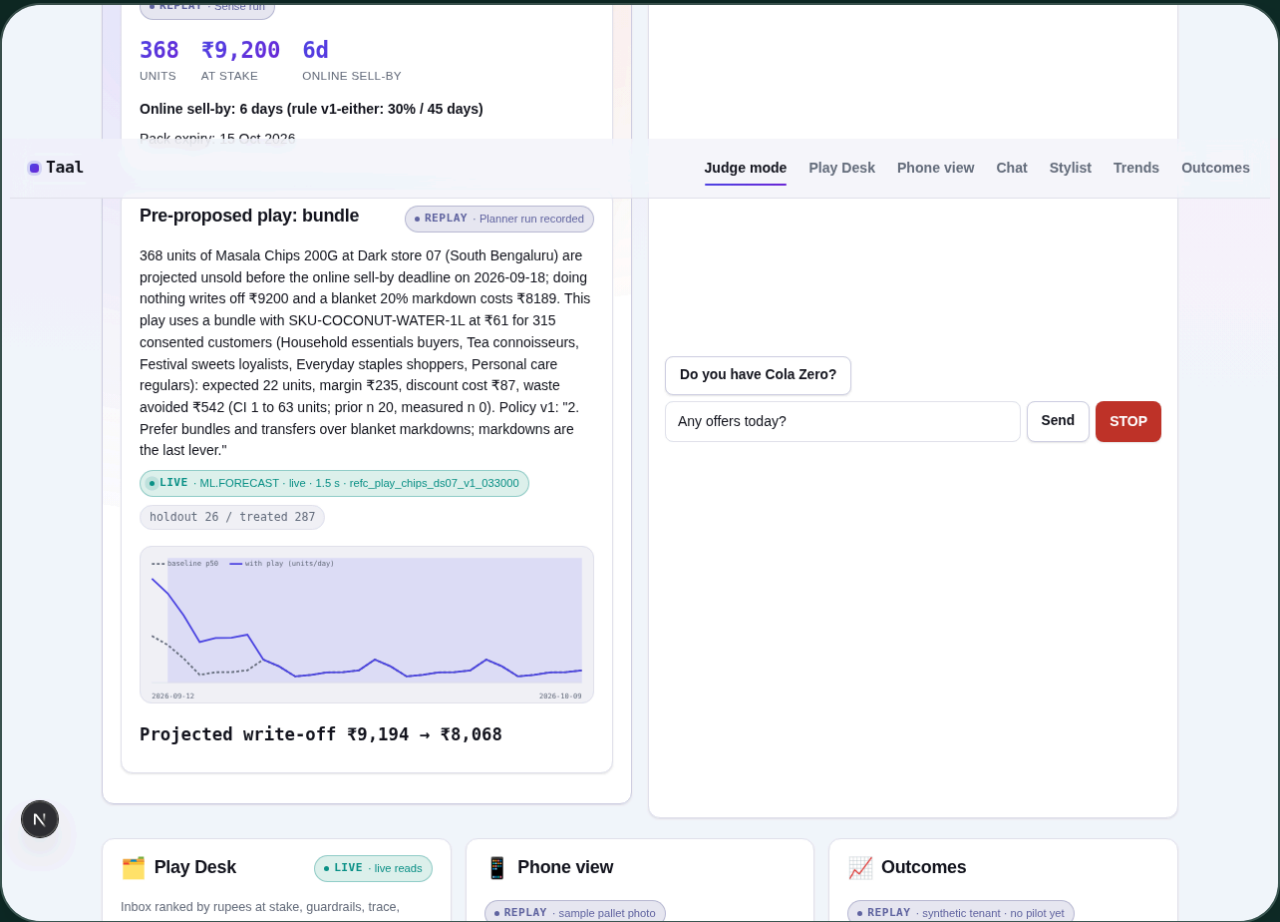


Taal

The agent that sells what the forecast says
you'll throw away.

Kutumb Mart · 10 dark stores, 6 outlets, Bengaluru

taal-web-2obkp776ca-el.a.run.app



WHAT IT DOES

Six steps from a forecast to a sale

1 · Capture

A node manager photographs a pallet; dates and counts are read with confidence scores, low-confidence rows confirmed before they reach inventory.

2 · Sense

Nightly per-SKU, per-node forecasts flag gaps — write-off risk, stockouts, unmet demand — with the rupees at stake and the deadline.

3 · Plan

An ADK Planner Agent designs a play; eight deterministic guardrails gate it; a failed guardrail sends it back for revision.

4 · Approve

A human approves; a holdout is assigned; the play is written into the forecast as a covariate and the series re-forecasts on the spot.

5 · Engage

A chat agent grounded in real stock and consent delivers the offer or an in-stock substitute; orders run through an order tool.

6 · Measure

Treated vs holdout, with a confidence interval; the estimator updates from the real result, so the next play starts smarter.

ONE REAL RUN, THIS TENANT ₹9,194 → ₹8,068 projected write-off on one lot, before and after a live guardrail-approved play

THE HOOK

33 days on the pack. 6 days to sell it online.

India's food regulator requires 30% of shelf life or 45 days remaining at delivery for food sold online. No forecasting tool knows that deadline exists — Taal computes it per lot, versioned as a tenant policy, and shows it on every gap card.

Demo lot, dark store DS-07

368 units

Masala Chips · ₹9,200 at stake

INNOVATION

One Play schema, two different problems

A **Play** is one object — target, mechanic, audience, guardrails, holdout — and the gate that checks it (8 guardrails plus the estimator) has zero code specific to either domain below. The same machinery runs both.

GROCERY

A forecast says a lot will go unsold before a deadline

5 gap types: online sell-by, expiry, stockout, rebalance, slow mover

APPAREL

A real customer asks for a style the node doesn't stock

2 gap types: unmet demand, assortment gap — sourced from real chat asks, not a forecast

Both land on the same Planner, the same 8 guardrails, the same holdout-measured loop — see Architecture for how a Play moves through it.

TECHNICAL MERIT

Real services, measured live

Sense (nightly, local model)

BigQuery ARIMA_PLUS_XREG SQL written, fixed, verified on one real series → gaps, segments, substitutes



Planner (ADK LoopAgent, Gemini Flash)

6 tools, max 3 iterations, revises on guardrail failure



Approve (Cloud Run)

holdout assignment, writes future_regressors, re-forecasts



Customer / Stylist Agent + Measure

Gemini chat, MCP order tool, treated-vs-holdout lift

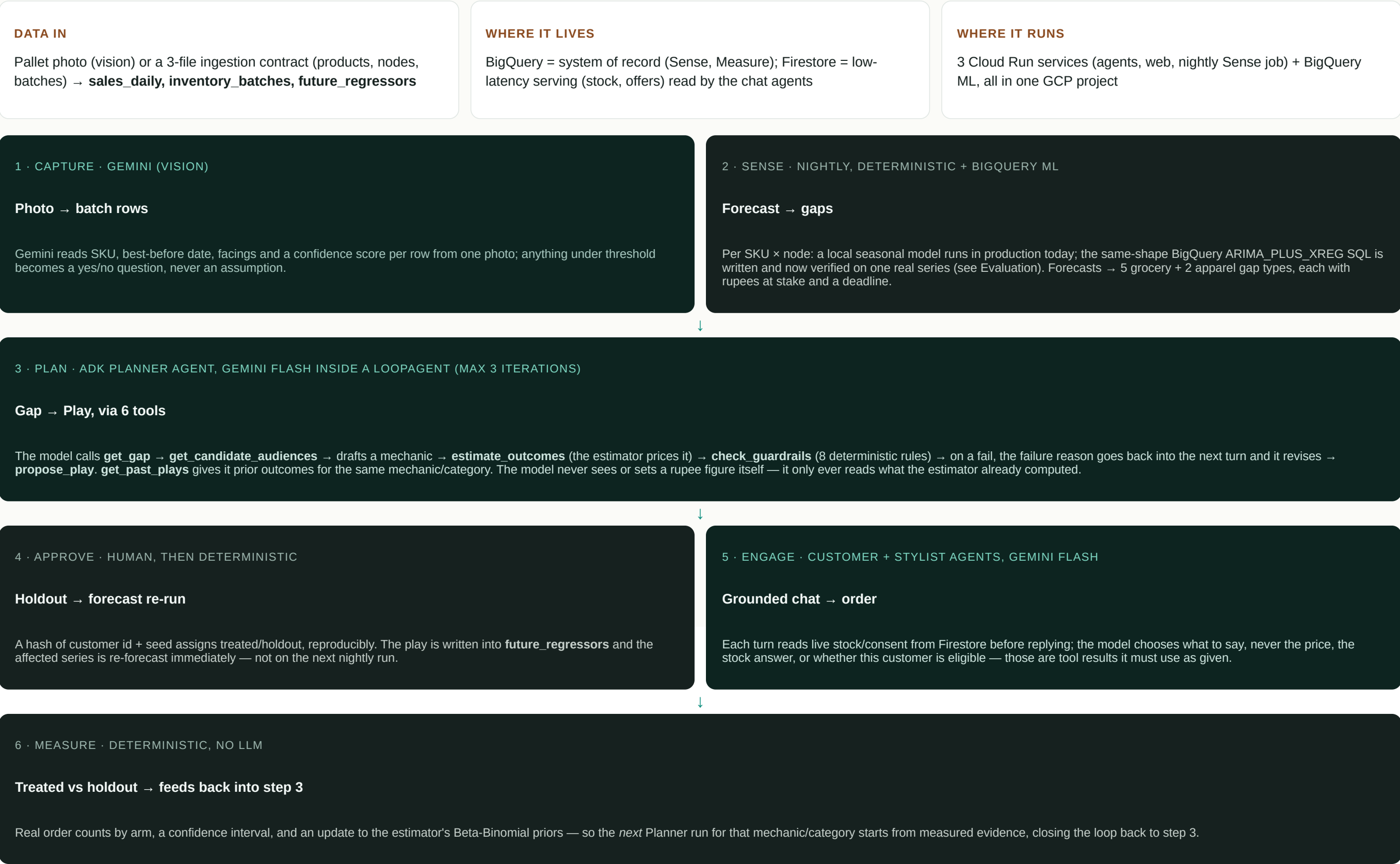
Measured, live Vertex — not estimated

Path	Latency
Customer chat, p50	4.40 s
Customer chat, p95	5.75 s
Approve → re-forecast	8.30 s
Sense, 300 SKUs × 16 nodes	3.83 s

50 real chat calls against Gemini 2.5 Flash on Vertex, no caching, a fresh session per call. Full raw output committed to the repo, not typed by hand.

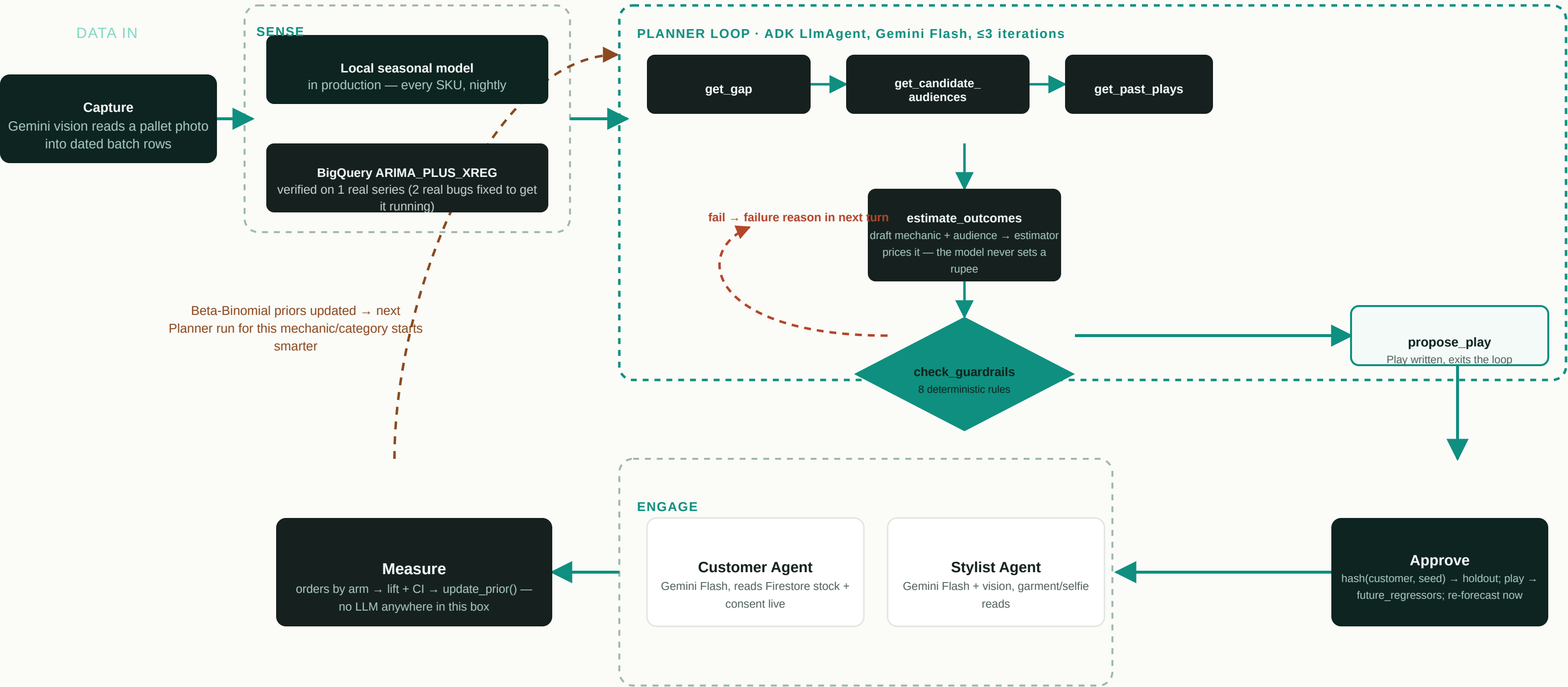
ARCHITECTURE

How data, agents and models fit together



ARCHITECTURE

System and agent orchestration, one lot's path through both



What Gemini decides — and what it is never allowed to

Gemini decides

- Which mechanic and audience for a gap, and the rationale
- Whether to revise a draft after a failed guardrail
- Reading a pallet photo into SKU, date, count, confidence
- Reading a garment or selfie photo into style attributes
- Dialogue: substitution reasoning, the reply envelope
- Vernacular copy for approved plays (falls back to templates)

Deliberately never LLM

- Every rupee — the estimator, a pure function
- All 8 guardrails — margin floor, consent, holdout, ...
- Holdout assignment — a hash, reproducible from seed
- Stock, offer eligibility, order prices
- Lift, confidence intervals, measurement

The trace panel shows this live: a guardrail returning **passed: false**, and the exact next tool call the Planner makes to fix it — not a black box.

INNOVATION

What makes this a decision engine, not a promo bot

A legal deadline as a model input

The FSSAI online sell-by rule is computed per lot and, once a play is approved, written straight into the forecast as a covariate — the series re-forecasts on the spot, not on the next nightly run.

One schema, two domains

Grocery write-off risk and apparel unmet-demand run through the identical Play object, the same eight guardrails and the same estimator — zero domain-specific code in the gate layer.

An estimator that learns

Every Measure run updates the estimator's priors from real treated-vs-holdout counts, so the next play of that shape starts from evidence, not a static assumption.

Not a fourth marketing channel

A decision layer: it emits an audience, copy and price a retailer's existing CRM or channel can send. It decides supply-side, per lot × node × deadline — not campaign scheduling.

Grounded, not general chat

The Customer Agent is grounded in this node's real stock and consent; it never promises a product the stock tool did not confirm.

PROBLEM ALIGNMENT & IMPACT

Three ways to handle the same lot

368 units of Masala Chips, dark store DS-07, 6 days to online sell-by — the same lot from the opening slide

DO NOTHING ₹9,200 written off; estimator projection at plan time	BLANKET 20% MARKDOWN ₹8,189 discounted for everyone, including customers who'd have paid full price; estimator counterfactual	THIS PLAY, LIVE-APPROVED ₹9,194 → ₹8,068 targeted to 287 consented customers, real re-forecast, not simulated
Why now The FSSAI online sell-by rule already exists; quick-commerce and dark-store growth make the online window the binding constraint, not the shelf.	Consent-native, not a growth hack Every audience is filtered by marketing consent before a play exists; STOP withdraws it immediately, checked on every future play, not just at signup.	

Honest about the gap: this is the seeded demo tenant. No pilot with real customers has run yet — the Outcomes screen labels every row REAL PILOT or SYNTHETIC and shows **unmeasured** rather than guess a lift below the minimum sample size.

EVALUATION

Numbers we can defend

Metric	Result	Note
Agent Simulation, guardrail pass	190 / 190	live Vertex, 5 guardrails, 190 personas
Vision read accuracy, 30 photos	30 / 30	synthetic photos — not a real-camera test
Customer chat p50 / p95	4.40 s / 5.75 s	real gemini-2.5-flash, 50/50 live calls OK
Forecast backtest MAPE, 9 tiers	0.107–0.221	7 rolling origins, 63 rows
Sense throughput	3.83 s	300 SKUs × 16 nodes, one run
BigQuery ARIMA_PLUS_XREG, one series	28/28 rows	real job, 2 real SQL bugs found & fixed running it
Pilot: treated vs holdout lift	not yet measured	no pilot has run
Cost per play, billing export	not yet measured	list-price estimate only, labelled

9 of 13 §12 rows carry a real, committed measurement; the rest say so rather than guess. Every number above is parsed from raw output under **eval/raw/**, never typed by hand.

LONGEVITY

Scalability and sustainability

Throughput, measured

300 SKUs × 16 nodes senses in 3.83 s. Cloud Tasks fan-out and per-gap triage by rupees-at-stake are the extrapolation path to a 20,000-SKU retailer.

Unit economics

Flash-only, no idle endpoints, nightly batch. A Cost Governor gates which gaps get a model call at all, by rupees at stake.

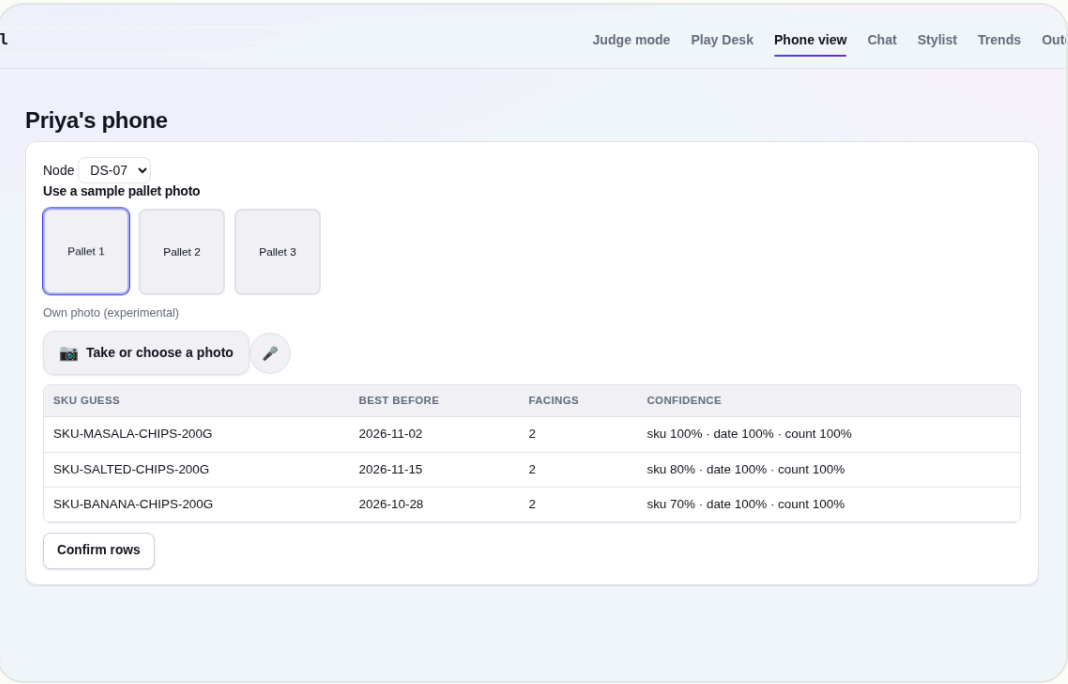
Onboarding

A three-CSV ingestion contract (products, nodes, batches) is the path from a seeded demo tenant to a real retailer's own catalogue.

Multi-tenant from day one

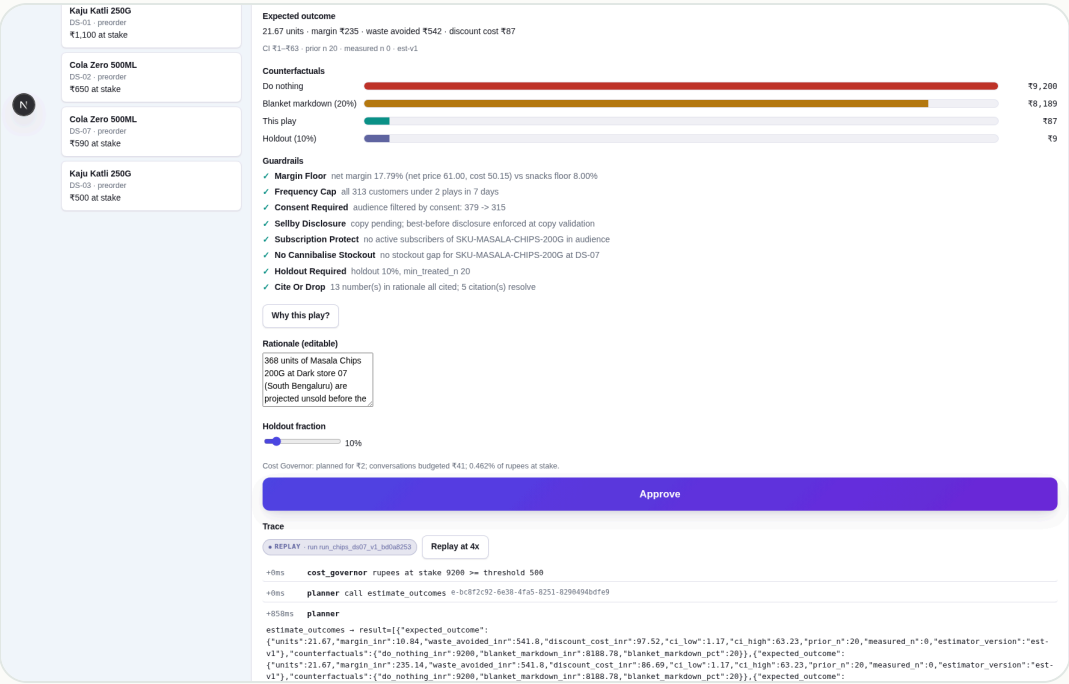
tenant_id on every table, per-tenant policy text — not a single-demo hack. Least-privilege IAM per service is written, not yet wired into the live deploy.

Three people, three screens, one job each



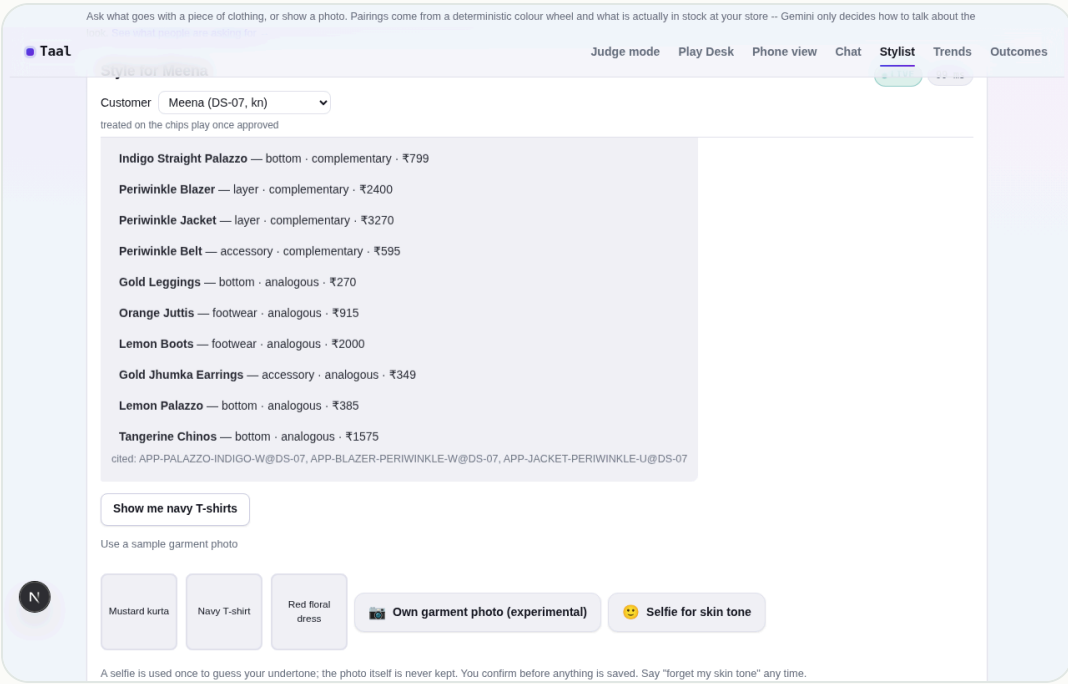
Priya, node manager

Photographs a pallet; low-confidence rows are confirmed, never assumed, before they reach inventory.



Arjun, demand planner

Reviews the rationale, the guardrails, the trace of a failed-then-revised draft, and approves.



Meena, customer

Gets the offer in her language with the best-before date, or an in-stock substitute, never a promise the stock tool didn't confirm.

CLOSING

Every forecast tells a retailer what will be thrown away.

Taal is the agent that sells it first.

REAL

Forecasts (local model; BigQuery SQL written and verified on one series), gaps, estimator, guardrails, the Planner loop, holdout assignment, live re-forecast, chat, orders, measurement

SIMULATED, DISCLOSED

The tenant Kutumb Mart (seeded generator, its rule stated in code); no pilot has run yet — the Outcomes screen says so on every row

taal-web-2obkp776ca-el.a.run.app

github.com/Nandish3010/probable-fortnight